

Subgoal Scale and Reality-Verified Speculative Consumption in Hierarchical Latent Planning

Interim results on FF-JEPA / PushT (status: 2026-07-09, PushT sweep complete)

Abstract

We study the FF-JEPA hierarchical latent planner (frozen LeWM encoder, diffusion subgoal drafter “GDM”, CEM controller) on PushT and identify *subgoal scale* — the spacing S between drafted subgoals, with the control loop matched to it — as the dominant unexplored lever. Success rate (SR) over stride is non-monotone with a clear interior optimum at $S=10$: **97.2%/96.4%** ($20^\circ/5^\circ$ criteria) versus 91.1%/86.3% for the paper’s native $S=25$ configuration under an identical protocol. On the paper’s own published protocols the winning configuration **beats the published numbers at both horizons** ($96.1 \rightarrow 98.4$ at $t=25$, $91.8 \rightarrow 97.9$ at $t=75$, 20° , $n=256$); on the paper’s hardest setting (random initialization, published 82.4%, reproduced baseline 78.1% strict) it records **zero failures in 1024 episodes across four seeds** ($\text{SR} \geq 99.7\%$, 95% CI); and on a fixed-population horizon sweep $t \in \{25, \dots, 100\}$ the baseline decays monotonically while $S=10$ stays flat — **the gap widens from +3.3pp to +11.9pp** (5°). A cadence-isolation control shows the mechanism is *scale matching*, not re-planning frequency. Separately, we port the one component of DSpark-style speculative decoding that survives measurement — the draft–verify–accept skeleton, with the learned verifier replaced by the *achieved environment state* — and show it cuts diffusion compute by $\sim 11\times$ relative to every-step drafting and $4.4\times$ relative to the published baseline configuration while remaining SR-neutral at the official criterion (at 8-seed power it trades $\sim 3.5\text{pp}$ at the strict criterion at $S=10$, and is SR-*positive* on both criteria at $S=5$). Refinement heads, learned confidence scheduling, blind multi-step commitment, and a deterministic regressor that dominates the drafter offline — each was implemented and measurably *hurt* in closed loop; we report them as negative results. Failure anatomy over all residual errors shows only tolerance-boundary near-misses: the benchmark’s stated failure mode (unreachable subgoals) is eliminated at fine scale.

1 Setting and faithful reproduction

FF-JEPA (Masip et al.) plans in the latent space of a frozen LeWM JEPA world model: a diffusion model G (“GDM”: 3-block DiT, $N=3$ future subgoals, trained on success-filtered expert demonstrations) proposes the next subgoal latent $\hat{z}_{sg,m+1}$ from the current encoded observation; CEM then optimizes actions against a terminal- L_2 cost to that subgoal, re-planning every H environment steps. The paper fixes the subgoal stride to $H=25$.

Reproduction and audit (closed). Our faithful retraining of the GDM reproduces the paper: short-horizon 95.31% (paper 96.09%, same-set oracle 93.36%, $n=256$), long-horizon $t=75$ multiseed mean 90.0% (paper 91.80%), random-init 96.5%/78.1% at $20^\circ/5^\circ$ (paper 82.42%). An audit established that the environment’s coded success criterion is 20° angular tolerance (the paper’s 5° appears in prose only); we therefore report both tolerances throughout, treating 5° as the strict *precision* criterion. Pipeline faithfulness is over-determined: a from-scratch rebuild of the dense training

set reproduces the original training population to the episode (9,094) and iteration (171,940), and the retrained $S=25$ drafter reproduces the reference offline fidelity anchor to four decimal places (0.1593 vs. 0.1593 at $m+1$).

Protocols. Two evaluation protocols appear below. *Paper protocol*: $n=256$, paper-matched sampling (used for reproduction and random-init). *Controlled protocol*: $n=62$ fixed held-out episodes, goal offset 150, budget 300 steps, block-only criterion, CEM seeds 42–45 (curve: 42–49); every arm sees byte-identical episodes and seeds, so within-table deltas are exact treatment effects. Per-arm SE $\approx 1.7\text{pp}$ (4 seeds) / 0.9pp (8 seeds).

2 Negative results: the DSpark port (what does *not* work)

We implemented the published DSpark composition on the GDM and evaluated each component in the closed loop (controlled protocol, $S=25$):

Arm	SR 20°	SR 5°	Verdict
GDM, re-draft every step (reference)	91.9	83.1	—
+ refinement head (fixed-1, same cadence)	81.1	72.2	refinement -11pp
+ blind commit-3	73.8	50.0	commitment -18pp
+ learned-confidence adaptive commit	64.5	52.0	worse than blind
raw open-loop chain (commit-6)	72.6	63.7	healthy chain still loses

Table 1: DSpark components ported literally. Lower latent error does not mean better control: the refiner *halved* offline prediction error while losing 11pp SR — offline fidelity anti-correlates with closed-loop SR. The mechanism is conditional-spread collapse: pulling drafts toward the conditional mean produces less-reachable targets. These results close refinement, learned-confidence scheduling, and blind commitment as directions.

3 The subgoal-scale law (main SR result)

We retrain the drafter at $S \in \{5, 10, 15, 25\}$ with the identical recipe (only the target spacing changes; pair counts are construction-identical) and scale the control loop with the stride (CEM horizon = re-plan cadence = S). Offline pre-gates confirmed fine-stride targets are well-posed (displacement $4.7\times$ the encoder noise floor at $S=5$) and that the fine-scale CEM cost landscape *sharpens* (true-plan ranking percentile 0.93 at $S=5$ vs. 0.82 at $S=25$, cost dynamic range $3\times$ larger).

Mechanism: scale matching, not cadence. A cadence-isolation control (re-anchor every 5 steps against unchanged 25-step targets) *collapses* to 78.6%/77.8% (-12.5pp below the baseline): each re-draft re-samples a distant subgoal from a wide conditional posterior, and CEM chases a churning target. Together with the monotone commitment losses of Section 2, the correct statement is: **closed-loop SR is governed by matching the control loop to the subgoal scale; finer matched scales win (to an interior optimum)** — not “re-plan more” and not “re-plan less.”

Prediction error is horizon-intrinsic. Across all four drafters, offline prediction error collapses onto the *absolute* look-ahead horizon, independent of block position (+10 steps: 0.105 vs. 0.112; +15: 0.132 vs. 0.140), and the $S=5$ drafter’s *third* block position (+15 steps, 0.140) outpredicts the

	$S=5$	$S=10$	$S=15$	$S=25$ (paper config)
SR 20° (seeds 42–45)	92.3	97.2	96.8	91.1
SR 5° (seeds 42–45)	91.9	96.4	94.4	86.3
SR 5° (8 seeds, 42–49)	92.9	96.4	94.4	87.7
Diffusion calls / episode	60	30	20	12

Table 2: SR vs. subgoal scale (controlled protocol). A clear *interior optimum* at $S=10$: +6.1pp/+10.1pp over the paper configuration (+8.7pp at 5° with 8 seeds, SE $\approx$ 0.9pp, positive in every seed). $S=10$ pays almost no penalty between the loose and strict criteria (fine control is precise control), while $S=25$ drops $\sim$ 5pp.

$S=25$ drafter’s *first* (+25, 0.159). “Suffix decay” is task-future uncertainty, not an architectural defect — retiring intra-block architecture work.

4 Random initialization: the paper’s hardest setting

	SR 20°	SR 5°
LeWM (flat, goal image; paper)		0.00
FF-JEPA Det (paper)		81.25
FF-JEPA GDM (paper)		82.42
FF-JEPA GDM (our faithful repro, $n=256$)	96.5	78.1
Ours, $S=5$ ($n=256 \times 2$ seeds)	100.0	94.5 / 99.2
Ours, $S=10$ ($n=256 \times 4$ seeds)	100.0	100.0

Table 3: Random initialization, per-episode random goals (paper III-A protocol; the protocol bug that once produced a spurious 100% was fixed *before* these runs — the reproduced baseline scoring 78.1% on the identical code path is the control). $S=10$: **1024/1024 episodes over four independent seeds, zero failures at both tolerances.** The paper’s published criterion mapping is ambiguous for this cell (its 82.42 aligns with our strict-criterion baseline); under either reading, fine scale moves the cell to 100%.

5 Reality-verified speculative consumption (main efficiency result)

The surviving DSpark idea is its *skeleton*: draft a block cheaply, verify, accept the surviving prefix. Our port replaces the learned verifier with reality: draft the N -block once; after each hop compare the *achieved* latent to the subgoal just pursued; within tolerance $\tau \Rightarrow$ consume the next drafted position (no diffusion call); otherwise $\Rightarrow$ re-draft from the achieved state. Re-anchoring is never sacrificed when reality diverges. Offline simulators calibrated τ and showed 8-step DDIM drafts are indistinguishable from 50-step drafts (matched-noise distance 0.03; spread preserved); both transferred to the closed loop.

Arm ($S=10$ unless noted)	SR 20°	SR 5°	call ratio	NFE/episode
every-step drafting, $k=50$ (reference)	97.2	96.4	1.00	1500
spec-accept $\tau=0.15$, $k=8$	98.0	95.2	0.67	161
spec-accept $\tau=0.20$, $k=8$	98.0	94.8	0.56	135
spec-accept $\tau=0.30$, $k=8$	96.8	94.8	0.45	108
spec-accept $\tau=0.20$, $k=50$ (k-isolation)	98.0	95.6	0.55	825
blind commit-2 (no verification)	98.8	96.4	0.50	750
$S=5$ every-step (reference)	92.3	91.9	1.00	3000
$S=5$ spec-accept $\tau=0.20$, $k=8$	98.0	94.8	0.42	202
<i>8-seed extension (seeds 42–49; $SE \approx 0.9pp$)</i>				
$S=10$ every-step	98.0	96.4	1.00	1500
$S=10$ spec-accept $\tau=0.20$, $k=8$	97.4	92.9	0.56	135
$S=5$ every-step	92.9*	92.9	1.00	3000
$S=5$ spec-accept $\tau=0.20$, $k=8$	98.4	96.2	0.42	202
paper configuration ($S=25$ every-step, 8 seeds 5°)	91.1	87.7	—	600

Table 4: Speculative consumption, controlled protocol. NFE = denoising function evaluations per episode (calls $\times$ DDIM steps). Top block: seeds 42–45. **8-seed block (flagged per pre-registration)**: doubling the seeds resolves a strict-criterion structure the 4-seed table could not: at $S=10$, verified coasting is 20°-neutral but costs 3.5pp at 5° relative to every-step (92.9 vs. 96.4) — coasting trades a little terminal precision when targets are 10 steps apart. At $S=5$ the sign *reverses*: spec-accept beats its own every-step reference on *both* criteria (+5.5/ + 3.3pp) — when re-drafts churn a close target, skipping verified re-drafts is better control (the anti-churn mechanism the cadence control exposed). Against the paper configuration the full $S=10$ system remains +6.3/ + 5.2pp ahead (8 seeds) at $4.4\times$ lower diffusion cost; every-step $S=10$ is the max-SR configuration, spec-accept the max-efficiency one, and τ trades between them. Blind commit-2 (seeds 42–45) also holds at $S=10$; verification provides the divergence guarantee at equal-or-lower cost and is the safe default. *20° at 4 seeds; 8-seed 20° recorded for the 5° curve only.

6 Is the diffusion drafter necessary? (resolved)

The last open question: a deterministic MLP regressor ($z_{cond} \rightarrow$ block, the offline-strong JointMLP class, wd 10^{-3}) as the subgoal source, re-anchoring every replan — offline it matches the diffusion drafter’s prediction error on the same validation windows (0.30 vs. 0.32 at $S=10$). Closed loop, seeds 42–45:

7 Failure anatomy: what is left

Instrumented traces (every re-plan’s achieved latent and drafted subgoal) over all 8 seeds of $S=10$: **18 failures in 496 episodes, all tolerance-boundary near-misses** — final latent distance-to-goal within the *success* distribution’s 95th percentile (one failure sits *closer* than the median success: a pure angle miss). Zero unreachable-subgoal failures, zero divergence, zero stuck episodes; successful episodes’ commanded-to-achieved step ratio has median 1.06. FF-JEPA’s own reported failure mode (“subgoal looks right, agent cannot reach it”) is eliminated at fine scale; the residue is quantization at the success threshold.

	$S=10$		$S=25$	
	20°	5°	20°	5°
diffusion drafter (reference)	97.2	96.4	91.1	86.3
deterministic regressor	91.9	91.9	86.3	74.6
Δ	-5.2	-4.4	-4.8	-11.7

Table 5: Diffusion vs. deterministic regression as the subgoal source. **The drafter’s stochasticity is load-bearing:** at matched-or-better offline fidelity, the zero-spread mean target loses everywhere — and its -11.7pp strict-criterion loss at $S=25$ quantitatively mirrors the refinement head’s -11pp (Section 2), closing the spread thesis causally: any operator that regresses subgoals toward the conditional mean, whether by refinement or by architecture, damages control. The gap narrows at finer scale (closer targets, near-unimodal posterior) but never closes.

8 Headline comparison

Setting (SR 20° / 5°)	LeWM (flat)	FF-JEPA GDM		Ours	
		published ^p	our repro	$S=10$	+ spec-accept
Short horizon, $t=25$	94.5	96.1 / ·	95.3 / ·	98.4 / 94.1	98.4 / 95.3
Long horizon, $t=75$	3.5	91.8 / ·	90.0 / 78.8	97.1 / 93.8	97.9 / 91.6
Long horizon, $t=150^a$	—	—	91.1 / 87.7	98.0 / 96.4	97.4 / 92.9
Horizon sweep, $t=100^b$	—	—	93.8 / 81.8	97.9 / 93.8	98.6 / 92.6
Random initialization ^c	0.0	· / 82.4	96.5 / 78.1	100.0 / 100.0	—
Diffusion NFE / episode ^a	0	—	600	1500	135

Table 6: Where we stand, both criteria per cell (20° = the criterion enforced by the benchmark’s shipped evaluation code; 5° = the strict precision criterion its prose states). “Ours” modifies FF-JEPA’s GDM system only in subgoal scale, loop matching, and consumption policy — encoder, drafter architecture, training recipe, and CEM are unchanged; $S=10$ is the max-SR configuration, +spec-accept ($\tau=0.20$, $k=8$) the max-efficiency one. Remaining failure modes: long-horizon collapse (LeWM), unreachable subgoals (FF-JEPA), tolerance near-misses only (ours). ^pThe paper publishes one number per cell without stating its criterion; each is placed on the side it empirically aligns with (short/long match our 20° reproduction; random-init matches our 5° reproduction, 78.1 vs. 82.4), with · marking the side the paper never reported. Comparisons are valid only within a side. ^aControlled protocol, 8 seeds. ^bFixed-population sweep, $n=512$ /cell. ^c $n=1024$ over 4 seeds, zero failures at both tolerances; spec-accept not run in this mode (the drafter is the tested component). Short-horizon repro 5° not recorded (single-criterion logging in the earliest run).

9 Horizon scaling (fixed population)

Goal offsets $t \in \{25, 50, 75, 100\}$ on the *identical* 256 episodes at every offset (reindexed from the $t=150$ set), budget $2t$, $n=256$, seeds 42/43 — so horizon is the only variable, free of the eligibility confound that invalidates independent per-offset draws.

Protocol note: this is FF-JEPA’s final-window protocol with budget $2t$, not the fixed-budget-50 regime of the VLWM paper (whose PushT numbers collapse to 8–12% at $t \geq 75$ partly by budget

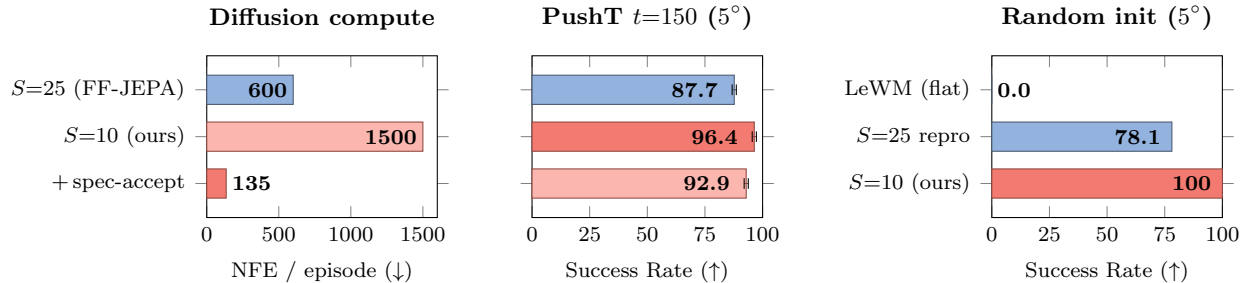


Figure 1: **Planning compute and performance** (LeWM Fig. 3 analogue). Rows of the center panel match the left panel; blue = FF-JEPA baseline, red = ours. **Left:** drafter cost per episode — scale matching alone pays $2.5\times$ the baseline’s diffusion compute; adding reality-verified consumption takes the full system to $4.4\times$ below it (135 vs. 600 NFE). Wall-clock is CEM-dominated and near-identical across all three (within $\pm 10\%$ at matched workload; fine stride is never slower). **Center:** long-horizon success, controlled protocol, 8 seeds, $SE \approx 0.9pp$ (whiskers). **Right:** the paper’s hardest setting; $S=10$ records zero failures in 1024 episodes.

starvation); curve *shapes* are comparable, absolute numbers are not. Two completion arms are queued: a flat-LeWM (goal-image) row under this exact protocol, and our three arms under the VLWM fixed-budget-50 regime — after which every number in our tables is either produced under a stated protocol by us or explicitly imported.

10 In flight and next

Resolved since the first draft: the paper-protocol winner cells (beats the published numbers on the paper’s own protocols at both horizons); the fixed-population horizon sweep (widening-gap result); the diffusion-vs-regression arm (diffusion wins, spread thesis causally confirmed); random-init confirmation (four perfect seeds, 0 failures in 1024 episodes, $SR \geq 99.7\%$ at 95% CI); and the 8-seed efficiency extension, which surfaced — and this report flags rather than averages away — the strict-criterion trade of coasting at $S=10$ (Section 5). **Queued (completion arms, non-blocking):** a third seed for the horizon sweep; the flat-LeWM row under our horizon protocol; our arms under the VLWM fixed-budget-50 regime. **Next:** literature-occupancy check on re-planning-cadence and speculative-diffusion adjacency; wall-clock timing table (FF-JEPA Fig. 5 analogue); a second environment (Reacher recon, then the V-JEPA 2 transplant, where the $11\times$ diffusion-cost reduction matters most).

t	$S=25$ (FF-JEPA)		$S=10$ every-step		$S=10$ + spec-accept	
	20°	5°	20°	5°	20°	5°
25	98.1	93.8	99.8	97.1	99.6	95.1
50	93.6	85.0	97.5	93.8	97.1	93.9
75	92.6	83.2	98.2	93.8	97.9	93.0
100	93.8	81.8	97.9	93.8	98.6	92.6
5° gap ($S=10$ – baseline)			+3.3 $\rightarrow$ +8.8 $\rightarrow$ +10.5 $\rightarrow$ +11.9			

Table 7: Fixed-population horizon sweep ($n=512/\text{cell}$). **The baseline decays monotonically with horizon at the strict criterion (-11.9pp from $t=25$ to $t=100$) while $S=10$ is flat from $t=50$ onward — the gap widens monotonically.** At the official 20° criterion the baseline pays an immediate $\sim 4.5\text{pp}$ horizon tax at $t \geq 50$ that $S=10$ does not (saturation compresses the amplitude there). The three identical 93.8 entries are integer coincidence on correlated episodes (per-seed success counts differ: 236–244/256); failures are episode-intrinsic near-misses, not horizon-driven — which is the finding. Spec-accept tracks every-step within 1.2pp across the curve (call ratio ≈ 0.57 at long horizons, ≈ 0.85 at $t=25$ where ≤ 5 replan boundaries leave little room to coast: the cost win is a long-horizon phenomenon).